

End report

Instructions for the teaching assistant

Implemented optional features

No optional features

Instructions for examiner to test the system

- Development platform
 - OS: Windows 11
 - CPU: AMD Ryzen 9 5900x
 - Docker version 27.0.3, build 7d4bcd8
 - Docker Compose version v2.28.1-desktop.1

Description of the CI/CD pipeline

- Version management
 - Just used the project branch
- Building tools
 - Gitlab CI/CD, Docker, Docker compose
- Testing
 - Tools
 - Pytest, requests
 - Cases
 - INIT state
 - PAUSED state
 - RUNNING state
 - Run-log
 - SHUTDOWN
- Packing
 - I used Docker images
- Deployment
 - Build phase
 - Docker-compose build
 - Test phase
 - Installs packages
 - Sets up network
 - Runs test
 - Docker-compose down
 - Deploy
 - Docker-compose down --remove-orphans
 - Docker-compose up -d

Example runs

This run failed because of SHUTDOWN test wanting a response

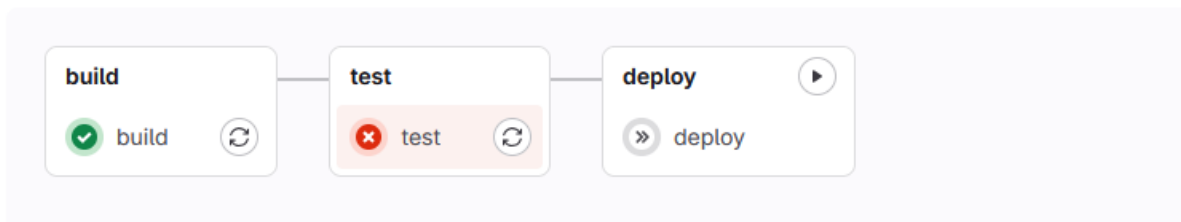
Fixes for networking

 **Failed** Jimi Niemi created pipeline for commit `8e47b3b5`  20 minutes ago, finished 17 minutes ago

For `project`

 3 jobs  3 minutes 14 seconds, queued for 3 seconds

Pipeline Jobs **3** Failed Jobs **1** Tests **0**



```
1315 E           requests.exceptions.ConnectionError: ('Connection aborted.', RemoteDisconnected('Remote end closed connection without response'))
1316 venv/lib/python3.11/site-packages/requests/adapters.py:682: ConnectionError
1317 ===== short test summary info =====
1318 FAILED tests/test_api.py::test_put_shutdown_state - requests.exceptions.ConnectionError: ('Connection aborted.', RemoteDisconnected('Remote end closed connection without response'))
1319 ===== 1 failed, 4 passed in 10.30s =====
1320 Running after_script
1321 Running after script...
1322 $ docker-compose down
```

00:23

This run went through, although it's an old one.

Fix requirements

 parent `a433d518`

 No related branches found > [Branches containing commit](#)

 No related tags found

 No related merge requests found

 Pipeline **#3862** passed with stage  in 2 minutes and 21 seconds

Changes **2** Pipelines **1**

Status	 Passed  00:02:21  1 hour ago
Pipeline	Fix requirements #3862  project  <code>348fc811</code> 
Stages	
Actions	 

```
882 rootdir: /builds/dqjini/comp.se.140
883 collecting ... collected 2 items
884 tests/test_api.py::test_get_state PASSED [ 50%]
885 tests/test_api.py::test_set_state PASSED [100%]
886 ===== 2 passed in 0.07s =====
887 Running after_script
888 Running after script...
889 $ docker-compose down
890 time="2025-01-31T15:12:27Z" level=warning msg="network default: network.external.name is deprecated. Please set network.name with external: true"
891 Container compse140-api-gateway-1 Stopping
```

Main learnings

The biggest difficulty was definitely networking the docker containers. Had a lot of situations where tests would work locally, but not in the pipeline because of connection issues. I'd say that doubled the time I did this project for. So as a main learning I'd say to learn how to configure container networks.

I also got to know that docker has pull limits during trying to fix the networking, which lead to not being able to test whether my pipeline goes through in the end. I got the error so close to the deadline that I couldn't wait the 6 hours, but I don't see a reason why it wouldn't pass.

Amount of effort used

I'd estimate the time used was somewhere around 25 hours.